

BAIO LEARNING SYSTEM

State PU Board • CBSE Class 11 • NEET-UG Biology Excellence Series

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SCAN TO CONNECT / BAIO APP

CHAPTER 8: CELL — THE UNIT OF LIFE

Target Annual Board Exam Master Series • Weightage Analysis • VSA (1M), SA-I (2M), SA-II (3M), LA (5M) & Mandatory Diagram Guide

☆ Board Exam Blueprint & Mark Distribution Insights:

Chapter 8 typically carries **7 to 10 marks** in Annual PU Board / Class 11 theory papers. Most frequent high-weightage questions include: (1) Fluid Mosaic Model with schematic diagram, (2) Detailed ultrastructure of Mitochondria or Chloroplast, (3) Morphological classification of chromosomes based on centromere position, (4) Prokaryotic cell envelope and mesosome functions, and (5) Differences between Gram-positive vs. Gram-negative bacteria or Plant vs. Animal cells.

PART A: VERY SHORT ANSWER QUESTIONS (VSA — 1 MARK EACH)

Q1. Who first observed and described a live, motile cell?

1 MARK

Anton von Leeuwenhoek (1674) first observed and described live cells under his improved microscope.

Q2. State the biological contribution of Rudolf Virchow to the Cell Theory.

1 MARK

Rudolf Virchow (1855) modified the cell theory by stating "**Omnis cellula-e cellula**", meaning all living cells arise strictly from pre-existing cells.

Q3. Name the smallest known living cellular organism and specify its dimensions.

1 MARK

Mycoplasma (PPLO), measuring approximately **0.3 µm** (0.1–0.3 µm) in length.

Q4. What is a plasmid? State its primary evolutionary significance in bacteria.

1 MARK

A plasmid is a small, extrachromosomal, circular double-stranded DNA molecule that replicates autonomously. It confers unique phenotypic traits such as **antibiotic resistance**.

Q5. What are mesosomes? Name any one vital function they perform.

1 MARK

Mesosomes are specialized membranous infoldings (vesicles, tubules, or lamellae) of the prokaryotic plasma membrane. Function: **Cellular respiration** / **DNA replication** / **Cell wall formation**.

Q6. Name the chemical substance that forms the middle lamella between plant cell walls.

1 MARK

Calcium and magnesium pectate (predominantly calcium pectate).

Q7. Which organelle is referred to as the "suicidal bag" of the cell, and why?

1 MARK

Lysosome, because it contains powerful acid hydrolases that digest cellular components and autolyze damaged or dying cells.

Q8. What is the sedimentation coefficient of ribosomes found in eukaryotic cytoplasm vs. mitochondria?

1 MARK

Eukaryotic cytoplasm contains **80S** ribosomes, whereas the mitochondrial matrix contains prokaryotic-like **70S** ribosomes.

Q9. Define tonoplast. What is its characteristic functional capability?

1 MARK

Tonoplast is the semi-permeable limiting membrane enclosing the plant central vacuole. It actively pumps ions and solutes **against concentration gradients** into the vacuolar lumen.

Q10. What are polysomes (polyribosomes)?**1 MARK**

A chain of multiple ribosomes attached simultaneously to a single mRNA strand to translate multiple copies of the same polypeptide.

PART B: SHORT ANSWER QUESTIONS (SA-I — 2 MARKS EACH)**Q11. State the two core postulates of modern Cell Theory.****2 MARKS**

1. All living organisms are composed of cells and products/secretions of cells (Schleiden & Schwann).
2. All cells arise from pre-existing living cells via cellular division — *Omnis cellula-e cellula* (Rudolf Virchow).

Q12. Differentiate between Gram-positive and Gram-negative bacteria based on cell wall composition.**2 MARKS**

Diagnostic Feature	Gram-Positive Bacteria	Gram-Negative Bacteria
Peptidoglycan Layer	Thick, multilayered (20–80 nm), retains crystal violet (purple).	Thin, single layer (8–10 nm), loses primary stain; counterstains pink/red.
Outer Membrane & LPS	Absent; contains teichoic acids.	Present; rich in lipopolysaccharides (LPS) and porins.

Q13. Differentiate between Rough Endoplasmic Reticulum (RER) and Smooth Endoplasmic Reticulum (SER).**2 MARKS**

- **RER:** Possesses 80S ribosomes bound on its outer surface via ribophorins; primary site for the synthesis and co-translational transport of **secretory proteins and lysosomal enzymes**.
- **SER:** Free of ribosomes, composed of branching tubules; primary site for **lipid, phospholipid, and steroid hormone synthesis**, as well as xenobiotic detoxification.

Q14. Distinguish between Cilia and Flagella in eukaryotic organisms.**2 MARKS**

- **Cilia:** Numerous, short, hair-like appendages operating in coordinated rhythmic waves like oars to move either the cell or surrounding fluid (e.g., *Paramecium*, ciliated epithelium).
- **Flagella:** Fewer in number (typically 1–4), significantly longer, exhibiting independent undulatory motion that propels the entire cell (e.g., *Euglena*, human spermatozoa).

Q15. Name the three types of plastids classified based on stored nutritional reserves.**2 MARKS**

Leucoplasts are non-pigmented nutrient-storing plastids classified into:

1. **Amyloplasts:** Specialize in storing starch / carbohydrates (e.g., potato tubers).
2. **Elaioplasts:** Specialize in storing fats, oils, and lipids (e.g., castor seeds).
3. **Aleuroplasts (Proteinoplasts):** Specialize in storing proteins (e.g., maize aleurone layer).

Q16. What are inclusion bodies in prokaryotes? Give two examples.**2 MARKS**

Inclusion bodies are non-membrane-bound reserve materials suspended freely within the prokaryotic cytoplasm. Examples: (1) **Phosphate granules** (volutin), (2) **Cyanophycean granules**, (3) **Glycogen granules**, and (4) **Gas vacuoles** in photosynthetic bacteria.

PART C: EXPLANATORY & ANALYTICAL QUESTIONS (SA-II — 3 MARKS EACH)

Q17. Explain the three-layered prokaryotic cell envelope and its functional significance.

3 MARKS

Prokaryotes possess a tightly bound three-layered envelope acting as a unified protective barrier:

1. **Glycocalyx (Outermost):** Polysaccharide coat present either as a loose *slime layer* (prevents water loss and desiccation) or a rigid *capsule* (protects pathogenic bacteria against phagocytosis by host white blood cells).
2. **Cell Wall (Middle):** Composed of rigid peptidoglycan (murein). Imparts mechanical shape and prevents the bacterium from bursting or collapsing under high internal osmotic turgor pressure.
3. **Plasma Membrane (Innermost):** Selectively permeable phospholipid bilayer governing ionic exchange, active solute transport, and enzymatic anchoring.

Q18. What is the Endomembrane System? Enlist its four component organelles and justify why Peroxisomes and Mitochondria are excluded.

3 MARKS

The endomembrane system consists of intracellular membranous organelles whose functions are **structurally and functionally coordinated**.

- **Constituent Organelles:** (1) Endoplasmic Reticulum, (2) Golgi Apparatus, (3) Lysosomes, and (4) Vacuoles.
- **Reason for Exclusion:** Peroxisomes, Mitochondria, and Chloroplasts are excluded because their biochemical functions (peroxide catabolism, cellular respiration, and photosynthesis) are completely autonomous and not coordinated with secretory vesicle sorting from the ER–Golgi pathway.

Q19. Describe the structural polarity and secretory functions of the Golgi apparatus.

3 MARKS

First described by **Camillo Golgi (1898)**, it consists of flattened parallel stacks of disc-like cisternae (0.5–1.0 μm).

- **Structural Polarity:** Exhibits two distinct interconnected faces:
 - *Cis face (forming face):* Convex surface facing the ER, receives newly synthesized transport vesicles.
 - *Trans face (maturing face):* Concave surface facing the plasma membrane, pinches off mature secretory vesicles.
- **Key Functions:** Post-translational modification, phosphorylation, glycosylation (formation of **glycoproteins and glycolipids**), and packaging of enzymes into primary lysosomes.

Q20. Why are Mitochondria and Chloroplasts termed "Semiautonomous Organelles"?

3 MARKS

Mitochondria and chloroplasts are termed semiautonomous because:

1. They possess their own **extranuclear circular double-stranded DNA genome** and distinct RNA transcripts.
2. They contain their own prokaryotic-like **70S ribosomes** capable of independently synthesizing a fraction of their structural and enzymatic proteins.
3. They divide autonomously within the host cell by **binary fission**, supporting the theory of endosymbiotic evolution.

Note: They are not fully autonomous because the majority of their mitochondrial/chloroplast proteins are encoded by the nuclear genome and imported from the cytosol.

Q21. Explain the internal axonemal ultrastructure of eukaryotic cilia and flagella (The 9 + 2 Pattern).

3 MARKS

Electron microscopy reveals that the axoneme (central core) is surrounded by the plasma membrane and exhibits:

- **Microtubular Array:** **9 peripheral doublets** of tubulin arranged circularly around **1 central pair of singlet microtubules**, generating the classic **9 + 2 axonemal array**.
- **Radial Spokes & Sheath:** The central singlet pair is enclosed by a central protein sheath, linked to one tubule of each peripheral doublet by 9 radial spokes.
- **Bridges & Dynein:** Adjacent peripheral doublets are interconnected by nexin linkers and bear inner and outer **dynein ATPase motor arms** that power sliding filament motility.
- **Basal Body:** Both cilia and flagella emerge from a centriole-like structure termed the **basal body** (which possesses a 9 + 0 triplet pattern).

PART D: COMPREHENSIVE LONG ANSWER QUESTIONS (LA — 5 MARKS EACH)

Q22. Explain the Fluid Mosaic Model of the Plasma Membrane proposed by Singer and Nicolson with its structural features and transport mechanisms.

5 MARKS

Proposed by **S.J. Singer and G.L. Nicolson (1972)**, this model is universally accepted as the authentic structural framework of biological membranes.

1. Structural Organization:

- **Quasi-Fluid Phospholipid Bilayer:** Composed of amphipathic phosphoglycerides with polar hydrophilic heads pointing outwards toward the aqueous medium and hydrophobic non-polar hydrocarbon tails tucked inside, protected from water.
- **Membrane Fluidity:** The quasi-fluid lipid matrix permits lateral movement of proteins within the bilayer. Fluidity is essential for cell growth, endocytosis, secretion, intercellular junctions, and cytokinesis.
- **Membrane Proteins:**
 - *Integral (Intrinsic) Proteins:* Partially or wholly embedded in the bilayer; many span the membrane as transmembrane proteins functioning as channels, pumps, or receptor permeases.
 - *Peripheral (Extrinsic) Proteins:* Lie on the outer or inner surface of the membrane and are easily dissociated by mild ionic treatment.
- **Carbohydrate Moiety:** Branched oligosaccharide chains attached to lipids (glycolipids) and proteins (glycoproteins) form the cell coat (glycocalyx), mediating cell–cell recognition and immune identity.

2. Transport Kinetics across Plasma Membrane:

- **Passive Transport:** Movement of neutral molecules along the concentration gradient ($C_{\text{high}} \rightarrow C_{\text{low}}$) without metabolic energy (ATP) expenditure. Water moves across by *osmosis*. Polar molecules traverse via carrier proteins by *facilitated diffusion*.
- **Active Transport:** Uphill movement of ions or solutes against the concentration gradient ($C_{\text{low}} \rightarrow C_{\text{high}}$) coupled to direct ATP hydrolysis (e.g., **Na⁺/K⁺-ATPase pump** transporting 3 Na⁺ outwards and 2 K⁺ inwards per cycle).

Q23. Describe the ultrastructure of a Mitochondrion with an analytical account of its compartmentalized enzymes and functions.

5 MARKS

Mitochondria are sausage-shaped or cylindrical organelles (diameter 0.2–1.0 μm, length 1.0–4.1 μm) visible under a light microscope only after staining with Janus Green B.

1. Structural Compartments:

- **Outer Membrane:** Smooth, continuous boundary containing pore-forming integral proteins called *porins*, rendering it freely permeable to small molecules, ions, and ATP.
- **Intermembrane Space (Perimitochondrial Space):** Compartment between outer and inner membranes containing enzymes such as adenylate kinase and cytochrome c.
- **Inner Membrane:** Selectively permeable barrier folded into numerous infoldings called **cristae**. Cristae significantly increase the catalytic surface area for ATP synthase complexes and electron transport chains.
- **Mitochondrial Matrix:** Dense, homogeneous colloidal ground substance enclosed by the inner membrane. Contains:
 - Complete enzymatic machinery for the **Krebs Citric Acid (TCA) Cycle**.
 - A single, circular, double-stranded covalently closed DNA molecule.
 - Prokaryotic-like **70S ribosomes** and tRNA pools for mitochondrial protein synthesis.
- **Oxysomes (F₀-F₁ Particles):** Pin-head structures studded on the inner membrane cristae. Consist of F₀ base (proton channel) and F₁ headpiece (catalytic ATP synthase synthesizing ATP via oxidative phosphorylation).

2. Functional Significance: Commonly termed the "**Powerhouse of the Cell**" because it is the exclusive site for aerobic respiration, beta-oxidation of fatty acids, and oxidative phosphorylation yielding ATP.

Q24. Classify chromosomes based on the position of the centromere. Describe their morphology and anaphase shapes.

5 MARKS

Every eukaryotic chromosome possesses a primary constriction termed the **centromere**, flanked laterally by disc-shaped trilaminar protein plates called **kinetochores** (attachment sites for spindle fibers).

Based on centromeric position, chromosomes are categorized into **four distinct morphological types**:

Chromosome Type	Centromere Location	Arm Ratio (p-arm / q-arm)	Anaphase Migration Shape
1. Metacentric	Strictly in the middle of the chromosome.	Two equal arms ($p = q$; arm ratio ≈ 1.0).	V-shaped during poleward anaphase pull.
2. Submetacentric	Displaced slightly away from the center.	One slightly shorter arm (p-arm) and one longer arm (q-arm).	L-shaped (or J-L transitional) configuration.
3. Acrocentric	Situated very close to one terminal end.	One extremely short, minute arm and one very long arm.	J-shaped during anaphase migration.
4. Telocentric	Located strictly at the extreme tip/terminal end.	Only a single arm; the short p-arm is completely absent.	I-shaped (rod-like) configuration.

Satellite Chromosomes (SAT-Chromosomes): Some chromosomes possess a non-staining, constant **secondary constriction** distal to the centromere. The small chromosomal fragment distal to this constriction is called a **satellite (trabant)**.

PART E: HIGH-YIELD DIAGRAMMATIC QUESTIONS (MANDATORY PRACTICE)

Q25. Mandatory Board Exam Diagram Practice Checklist

5 MARKS

In the Annual Board Examination, questions frequently state: "Draw a neat, labeled diagram of...". The top 5 diagrams from this chapter with their required labels:

1. Fluid Mosaic Model of Plasma Membrane:

Mandatory Labels: Phospholipid bilayer, Hydrophilic polar head, Hydrophobic non-polar tail, Integral (transmembrane) protein, Peripheral protein, Cholesterol, Glycoprotein, Glycolipid.

2. Longitudinal Section of Mitochondrion:

Mandatory Labels: Outer membrane, Inner membrane, Cristae, Intermembrane space, Matrix, F_0-F_1 ATP Synthase (Oxysome), Circular dsDNA, 70S Ribosomes.

3. Ultrastructure of Chloroplast:

Mandatory Labels: Outer membrane, Inner membrane, Intermembrane space, Stroma, Thylakoid, Granum (stack of thylakoids), Stroma lamella (fret channel), Thylakoid lumen, Stroma DNA, Plastoglobules.

4. Ultrastructure of Cilia/Flagella (Axoneme in Transverse Section):

Mandatory Labels: Plasma membrane, Peripheral microtubule doublets (9 pairs), Central singlet microtubules (2 singlets), Central sheath, Radial spoke, Interdoublet bridge (nexin), Dynein motor arms.

5. Types of Chromosomes Based on Centromere Position:

Mandatory Labels: Centromere (primary constriction), Kinetochores, Short p-arm, Long q-arm, Secondary constriction, Satellite region (SAT) in Metacentric, Submetacentric, Acrocentric, and Telocentric forms.